Machine Learning–ARIMA Framework for Stock Selection and Forecasting in the SL20 Index

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DECLARATION

I, B.M.Sachini Chamodya , declare that the presented project report titled, "Machine Learning–ARIMA Framework for Stock Selection and Forecasting in the SL20 Index" is uniquely prepared by me based on the group project carried out under the supervision of Dr.L.T.Wedage Department of Mathematics, Faculty of Science, University of Ruhuna, as a partial fulfillment of the requirements of the level III Case Study II course unit (MFM3151) of the Bachelor of Science Financial Mathematics and Industrial Statistics Degree in Department of Mathematics, Faculty of Science, University of Ruhuna, Sri Lanka. It has not been submitted to any other institution or study program by me for any other purpose.

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Date:.....

SUPERVISOR'S RECOMMENDATION

I/We certify that this study was carried out by B.M.Sachini Chamodya under my/our supervision.

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ABSTRACT

In this this study we developed Machine Learning–ARIMA framework for price forecasting and stock selection in the Colombo Stock Exchange's (CSE) SL20 Index.
The best performing stocks currently in CSE were selected based on market capitalization. In the first stage, Machine Learning techniques including classification, regression, and ranking models are employed to develop 3 analysis are fundamental, technical, and combined indicators to identify the best-performing company among selected SL20 firms. Then we found that the combine of both fundamental, technical indicators provides the best-performing model for forecasting the next 30 days of closing prices for the selected stock. Random Forest, XGBoost, and LightGBM ranking models were applied to rank ten selected SL20 companies, where Dialog Axiata PLC (DIAL) consistently ranked as the top-performing stock.
In the second stage time series techniques are used to analyze the future price behavior of the chosen stock. ARIMA(3,1,1) model is selected to predict DIAL's short-term price movement. The 30-day forecast reveals a stable upward trend with moderate uncertainty. Overall, this study demonstrates that integrating ARIMA forecasting with machine learning stock selection results in a reliable, data-driven tool to assist investors in the Sri Lankan stock market. .

Keywords: Machine Learning, ARIMA, Stock Selection, Price Forecasting, SL20 Index, Colombo Stock Exchange, Emerging Markets, Time Series Analysis

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CHAPTER 1
Introduction

1.1 Background of the Study

The financial market is complex and changes quickly because it is influenced by global and local economic factors. Traditional methods like fundamental analysis and technical analysis often cannot capture these complex relationships. This is especially true in emerging markets such as the Colombo Stock Exchange (CSE).

New advances in artificial intelligence and machine learning (ML) can help address these issues. ML can study large datasets and combine technical indicators .Time series models like ARIMA can also predict prices when the data is stationary and the model is set correctly. This study uses both ML based stock selection and ARIMA forecasting to build a strong decision-making framework for investors in Sri Lanka.

1.2 Significance of the Study

This research is useful for both academics and practitioners. For academics, it fills gaps in the literature by creating a framework that combines machine learning for stock selection with ARIMA for price forecasting. Such a framework is missing in current studies, especially for emerging markets. For practitioners, it gives investors, portfolio managers, and regulators linked to the Colombo Stock Exchange and SL20 index a data-based tool. This tool can help them make better investment decisions, work more efficiently, and predict returns more accurately in the Sri Lankan stock market.

CHAPTER 2
Literature Review

Investors, portfolio managers, and analysts want to find the best companies before they invest. Stock markets change very fast, so choosing strong companies helps reduce risk and increase returns. In Sri Lanka, many investors look at the SL20 Index because it includes the most active and financially strong companies on the Colombo Stock Exchange (CSE). But even inside the SL20, companies do not perform the same. Therefore, investors need good methods to find companies with strong financial health, steady growth, and good market performance. Recent studies show that machine learning and time series models are useful for this. Machine learning can group and rank companies using technical, financial, and market indicators. ARIMA models can predict future prices using past price data. When used together, these methods help investors make better decisions and choose high quality companies with strong future potential.

2.1 Machine Learning Approaches to Stock Selection

Many studies use machine learning to choose strong companies and predict stock prices. Research also shows that combining technical and fundamental indicators in one ML model is effective. For example, one study used moving averages, RSI, MACD, volatility, and volume along with PE, PB, dividend yield, EPS, and profitability ratios. They tested neural networks, decision trees, and random forests, and found that combining these indicators improves stock selection accuracy (1). Using both types of data helps because each reflects different parts of a company's performance.

Other studies focused more on fundamental indicators to identify financially strong companies. They used PE ratio, PB ratio, EPS, ROE, ROA, and other profitability measures. Machine learning models such as random forest, feed-forward neural networks, ANFIS, and feature-ranking techniques successfully ranked companies and identified the most important financial variables (2). This supports using fundamental indicators to evaluate SL20 companies. Some research used factor-based approaches, including weekly returns, volatility, momentum, size, value, and quality factors. Models like logistic regression, gradient boosting, random forest, and deep neural networks were able to classify

stocks as outperformers or underperformers with high accuracy (3). These studies also showed that risk-adjusted performance measures, like Sharpe-type ratios, help improve ML stock classification models.

More advanced frameworks combined large numbers of features. One study used 244 technical, fundamental, and market variables and applied models like logistic regression, random forest, and deep neural networks, enhanced with stacking ensembles and genetic algorithm-based feature selection (4). The results showed that combining multiple algorithms improves accuracy. Another study used 21 fundamental financial ratios (PE Ratio, EPS, ROE, Book Value) taken from company reports to predict how stocks will perform. Their methodology involved a comparative evaluation of a Feed Forward Neural Network (FNN) and an Adaptive Neuro Fuzzy Inference System (ANFIS) (5). Their results showed that the FNN model performed much better.

Some studies focused on value investing and quality measures. They used variables such as PE, PB, EV/EBITDA, ROE, ROA, revenue growth, cash flow growth, and profitability stability to classify companies and identify top performers using random forest and gradient boosting (6). Other research used hybrid ML approaches combining price, market, and categorical variables with models like linear regression, SVM, KNN, and random forest, applying PCA for dimensionality reduction (7). These studies show that ML models work best when using diverse financial indicators and accounting for company characteristics, which supports selecting the best SL20 company.

Overall, the literature shows that combining technical indicators, fundamental ratios, and market variables in ML models improves stock classification and selection. These findings strongly support using machine learning in the present study to identify the best-performing company among the ten selected SL20 firms.

2.2 Time Series Forecasting utilizing ARIMA Models

ARIMA is a widely used method for forecasting stock prices in financial studies. Many researchers have applied it to predict stock behavior. One study forecasted Nokia and Zenith Bank stock prices using the Box–Jenkins method and selected ARIMA(2,1,0) and ARIMA(1,0,1) as the best models based on stationarity tests, ACF/PACF plots, and

model fit criteria (12). The results showed high accuracy, proving that ARIMA works well for short-term prediction.

Another study used ARIMA to predict monthly prices of Solar Industries India Limited from 2000 to 2024. They chose ARIMA(0,2,1) based on AIC values and residual tests (9). They also found a moderate positive correlation between trading activity and stock prices, showing that ARIMA can capture long-term trends. Studies using daily prices also confirm ARIMA's usefulness. For example, Apple's daily closing prices over five years were analyzed, and ARIMA(1,1,1) was selected using AIC and Ljung–Box tests, with RMSE and MAPE measuring forecasting accuracy (10). Another study compared different ARIMA models, including auto ARIMA and ARIMA(1,1,33), using Netflix stock prices. ARIMA(1,1,33) gave the most accurate forecasts and showed stable future price behavior (11).

These studies show that ARIMA is effective for capturing linear price patterns and producing stable forecasts. This supports using ARIMA in the present study to forecast the stock price of the best-performing SL20 company selected through machine learning.

2.3 Research Gap and Problem Statement

Many studies have used machine learning to select stocks and ARIMA to forecast prices, but there are still gaps in research on the Sri Lankan market. Most studies focus on large markets such as the US, Europe, India, or China. Very few studies examine the Sri Lankan stock market, and almost none combine technical indicators, like moving averages, RSI, or MACD, with fundamental metrics, like PE ratio, PBV, EPS, or ROE. While this approach works well in other markets, it has rarely been tested for Sri Lankan companies.

Most local studies also rely on simple regression or basic technical indicators and do not use advanced machine learning or ensemble methods. This limits the accuracy of stock selection, and investors do not have systematic tools to rank companies or identify top performers in the SL20.

Another gap is that stock selection and price forecasting are usually treated separately. Studies either use machine learning to select stocks or ARIMA to forecast prices, but rarely both together. No research provides a framework that first ranks companies with

machine learning and then forecasts the top company's price with ARIMA. Without such an integrated approach, investors cannot make fully informed decisions.

These gaps show the need for a study that combines technical–fundamental machine learning models with ARIMA forecasting. This can help select the best-performing SL20 company and predict its future price, providing a more reliable and practical tool for investors in the Sri Lankan stock market.

Identified gaps:

1. **Geographical Gap (SL20 Companies):** Most studies use large global datasets from markets like the U.S., China, or Europe. Very few focus on Sri Lanka or a small set of companies. Research that applies ML models specifically to SL20 companies is limited.
2. **Methodological Gap (Selection vs. Forecasting):** Many studies either use ML to select stocks or use models like ARIMA to forecast prices. Only a few combine both methods, so investors do not have a complete framework that includes selection and forecasting together.
3. **Variable Gap (Emerging Markets):** Studies show that combining technical indicators (MA, RSI, MACD) with fundamental measures (PE, PBV, EPS, ROE) works well in global markets. However, almost no research tests this combined approach for SL20 companies in Sri Lanka.

CHAPTER 3

3.1 Problem Statement

The Sri Lankan stock market offers many financial, fundamental, and technical indicators, but investors lack a clear, data-driven method to find the best-performing company in the SL20 index. Most studies do not combine technical and fundamental variables, and almost none use machine learning to evaluate SL20 stocks. Moreover, no research provides a full framework that first selects the best company with ML and then forecasts its future price using time series models. Therefore, a structured ML-based selection and forecasting framework is needed to support informed investment decisions in Sri Lanka.

3.2 Aims and Objectives

The main aim of this study is to develop an integrated framework that identifies the best performing company among selected SL20 companies using machine learning techniques and forecasts its future stock price using ARIMA time series modeling.

3.2.1 Research Objectives

1. To identify the best-performing company in the CSE (SL20) using integrated technical and fundamental indicators applied via ML models.
2. To forecast the future stock price of the ML-selected company using Time Series (TS) models (ARIMA).

3.2.2 Research Questions

1. How effectively can Machine Learning models to identify the best-performing company among selected SL20 companies?
2. After selecting the best company using ML, how accurately can time series models forecast the future stock price of that company?

CHAPTER 4
Methodology

4.1 Research Design

This study has two main parts. First, Machine Learning models are used to find the best company from the selected SL20 stocks. Then, in the second part, the future stock price of that company is predicted using a suitable ARIMA time series model.

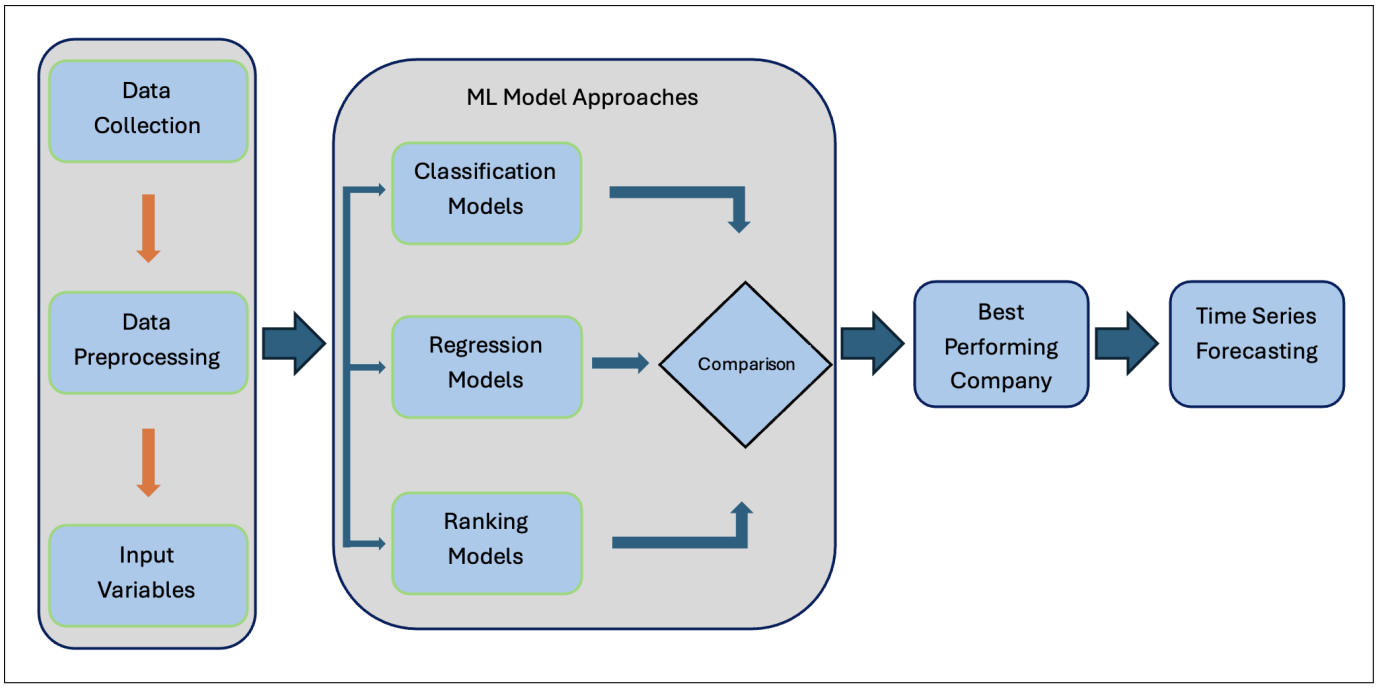


Figure 4.1: Proposed Research Framework.

4.2 Data

This study uses secondary data from the Colombo Stock Exchange (CSE) and TradingView. Ten SL20 companies were selected via CSE records, and three data types (fundamental, market price, and technical) were collected, following previous studies.

This study uses secondary data obtained from two main online sources:

- **Colombo Stock Exchange (CSE)**: `https://www.cse.lk`
- **TradingView**: `https://www.tradingview.com`

4.3 ML models approach

In this study applies seven machine learning techniques to select the best performing company among ten selected SL20 companies. All models use the same set of technical and fundamental variables to fair comparison.

4.3.1 Classification Models

Classification models are used to categorize stocks based on their expected performance in the next investment period.

Random Forest Classifier

Random Forest Classifier classifies SL20 stocks as outperforming or underperforming. It captures complex, non linear relationships between indicators and reduces overfitting by combining multiple decision trees. This helps identify stocks with strong potential for future performance.

XGBoost Classifier

XGBoost Classifier uses gradient boosting to classify stocks. Each new tree corrects errors from previous ones, improving prediction accuracy. It handles many financial indicators and estimates the probability that a stock will outperform others.

Support Vector Classification (SVC)

SVC separates outperforming and underperforming stocks by finding an optimal decision boundary. Using kernel functions, it captures non-linear relationships and remains robust in volatile and less efficient markets like the Colombo Stock Exchange.

4.3.2 Regression Techniques

Regression models are used to estimate expected future returns for each stock.

Random Forest Regressor

Random Forest Regressor predicts future stock returns by averaging results from many decision trees. It gives stable and reliable estimates, and stocks with higher predicted returns are considered stronger.

Support Vector Regression (SVR)

SVR predicts stock returns while reducing noise and extreme price effects. It focuses on overall trends and helps compare stocks based on their predicted performance.

4.3.3 Ranking Techniques

Ranking models directly order stocks based on relative performance.

XGBoost Ranker

XGBoost Ranker orders SL20 stocks from best to worst using a learning-to-rank method. It directly supports selecting the top stock, and the highest ranked stock is treated as the main candidate.

LightGBM Ranker

LightGBM Ranker efficiently ranks stocks using technical and fundamental indicators. It handles large data accurately and quickly, and the top-ranked stock is chosen for forecasting.

4.4 Comparative ML Analysis of Stock Approaches

The study implements three machine learning–based analytical frameworks to evaluate stock market performance:

1. **Fundamental Analysis Model**
2. **Technical Analysis Model**
3. **Combined (Fundamental and Technical) Analysis Model**

4.4.1 Fundamental Variable Analysis

Fundamental indicators describe a company's internal financial strength, growth, value, and risk. Ten main variables were selected from prior studies and grouped into three categories. Fundamental variables were obtained from `https://www.tradingview.com/` and are reported on a quarterly basis.

Target variable (Classification and Rnaking): Next 2-days future return.

- Positive value → Expected gain
- Negative value → Expected loss

Typical formulation:

- 1 → Stock is expected to deliver a high future return
- 0 → Stock does not deliver a high return

This target is used because it directly supports buy or not-buy investment decisions and allows probability-based predictions, which can be used to rank companies according to their likelihood of achieving future gains.

Earnings and Profitability variables:

These indicators show how well a company generates profit from its assets and shareholders' funds.

- **Earnings Per Share(EPS):** Profit earned per share. Higher EPS usually leads to higher stock prices.
- **Return on Equity(ROE):** Profit earned from shareholders' equity. A high ROE shows efficient use of invested capital.
- **Return on Assets(ROA):** Profit earned from total assets. A high ROA indicates good use of company resources.

Operational Efficiency and Growth variables:

These variables measure day-to-day performance and business expansion.

- **Gross Profit Margin:** Revenue remaining after direct costs. Higher margins show efficient operations.
- **Operating Cash Flow per Share(OCFPS):** Actual cash generated from operations. Strong OCFPS shows good liquidity.
- **Revenue Growth:** Increase in total sales over time. Higher growth signals business expansion.

Risk and Valuation variables:

These indicators reflect the company's financial risk and how the market values the firm.

- **Debt-to-Equity Ratio(D/E):** Shows leverage level. Higher values indicate higher financial risk.
- **Price-to-Earnings Ratio(P/E):** Market value per unit of earnings. Higher P/E reflects high growth expectations.
- **Price-to-Book Ratio(P/B):** Market price compared to net asset value. Helps identify under or over valued stocks.
- **Price-to-Sales Ratio(P/S):** Market value compared to total revenue. Useful when earnings are unstable.

4.4.2 Technical & Price-Based Variable Analysis

Technical and price-based variables describe stock price behaviour, market activity, and short-term trading patterns. These indicators are used to support both machine learning classification and ARIMA forecasting.

Target variable (Classification and Rnaking): Daily future return.

Typical formulation:

- $1 \rightarrow$ Stock is expected to deliver a high future return
- $0 \rightarrow$ Stock does not deliver a high return

Price and Volume Indicators

- **Closing Price (CP):** The main daily price used for calculating returns and technical indicators.
- **Opening Price (OP), Highest Price (HP), Lowest Price (LP):** Represent daily price movement and intraday volatility.
- **Trading Volume (VOL):** Number of shares traded; higher volume indicates stronger market participation.
- **Daily Return (RET):** Percentage change in price from one day to the next.

Price and Volume Indicators

- **MA50:** A 50-day moving average that reflects the overall price trend.

$$MA_{50}(t) = \frac{1}{50} \sum_{i=0}^{49} P_{t-i} \tag{4.1}$$

  where P_{t-i} is the closing price on day $t - i$.

- **Relative Strength Index(RSI):** Identifies whether a stock is overbought or oversold.

$$RSI = 100 - \frac{100}{1 + RS}, \quad RS = \frac{\text{Average Gain over } n \text{ periods}}{\text{Average Loss over } n \text{ periods}} \tag{4.2}$$

- **Beta(β):** Measures the stock's sensitivity to overall market movements (SL20).

$$\beta = \frac{\text{Cov}(R_s, R_m)}{\text{Var}(R_m)} \tag{4.3}$$

  where R_s and R_m are the stock and market returns, respectively.

- **Volatility:** Indicates how much the stock price fluctuates over time.

$$\text{Volatility} = \sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (R_i - \bar{R})^2} \tag{4.4}$$

  where R_i is the daily return and $\bar{R}$ is the mean return over n days.

4.4.3 Combine Anlysis

combined analysis integrates both fundamental and technical features, while keeping the same target variables to maintain methodological consistency.

Target variable (Classification and Rnaking): Next 2-days future return.

- Positive value → Expected gain
- Negative value → Expected loss

Typical formulation:

- 1 → Stock is expected to deliver a high future return
- 0 → Stock does not deliver a high return

4.5 Time Series Forecasting

Time series forecasting addresses a wide range of problems using various methods. After selecting the best company, an appropriate ARIMA based time series model will be applied to forecast its stock prices based on the company's data features.

Time Series Forecasting Models Based on ARIMA Model

ARIMA model is a widely used method in time series analysis, where the output variable is cross–correlated with a non-identical random variable.

- Auto-regressive (AR) Model
- Moving Average (MA) Model
- Auto-regressive Moving Average (ARMA) Model
- Auto-regressive Integrated Moving Average (ARIMA) Model
- Seasonal Auto-regressive Integrated Moving Average (SARIMA) Model

1.Autoregressive (AR) Model

AR models predict future values from past values, exemplified in stock price prediction based on past performance.

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t \tag{4.5}$$

Where:

- X_t: Current value of the time series.
- $\phi_1, \phi_2, \ldots, \phi_p$: Coefficients for past values.
- c: Constant term.
- ϵ_t: White noise error term.
- p: Order of the AR model (number of lags).

2. Moving Average (MA) Model:

Moving average models use past error terms for linear modeling of univariate time series. Versatile in technical analysis, they identify trends, support/resistance levels, and buying/selling opportunities based on current and past values.

Equation:

$$X_t = c + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \cdots + \theta_q\epsilon_{t-q} \tag{4.6}$$

Where:

- X_t: Current value of the time series.
- $\theta_1, \theta_2, \ldots, \theta_q$: Coefficients for past error terms.
- $\epsilon_t, \epsilon_{t-1}, \ldots$: White noise error terms.
- c: Constant term.
- q: Order of the MA model (number of lagged errors).

3.Autoregressive Moving Average (ARMA) Model

ARMA model combines AR and MA for effective linear modeling of stationary time series with minimal parameters for comprehension and forecasting.

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \cdots + \theta_q\epsilon_{t-q} \tag{4.7}$$

Where:

- $\phi_1, \phi_2, \ldots, \phi_p$: AR coefficients.
- $\theta_1, \theta_2, \ldots, \theta_q$: MA coefficients.
- p: Order of the AR component.
- q: Order of the MA component.

4. Auto-regressive Integrated Moving Average (ARIMA) Model

The Auto-regressive Integrated Moving Average (ARIMA) model is a widely used technique for time series forecasting. It combines three fundamental components to capture different characteristics of the data.

- **AR (AutoRegressive) Component**: The AR component models the relationship between the current observation and a specified number of its lagged values (past observations).
- **I (Integrated) Component**: The Integrated component involves differencing the original time series to achieve stationarity by removing trends or other non-stationary patterns.
- **MA (Moving Average) Component**: The MA component captures the relationship between the current observation and the residual errors from previous time steps.

The general form of the ARIMA model is denoted as ARIMA(p, d, q), where:

- p : Order of the autoregressive (AR)
- d : Degree of differencing required to make the time series stationary
- q : Order of the moving average (MA) term

4.6 Exploratory Data Analysis

Dialog Axiata PLC (DIAL), which was identified as the top-ranking stock through the machine learning selection process.To understand the fundamental characteristics of the Dialog stock price series, an Exploratory Data Analysis (EDA) was initiated, involving data cleaning, descriptive statistics, and visualization of historical price movements. The initial time series plot of Dialog's daily stock prices from 2015 to 2025 revealed a dynamic behavior characterized by an overall upward trajectory, though short-term fluctuations made direct interpretation challenging.

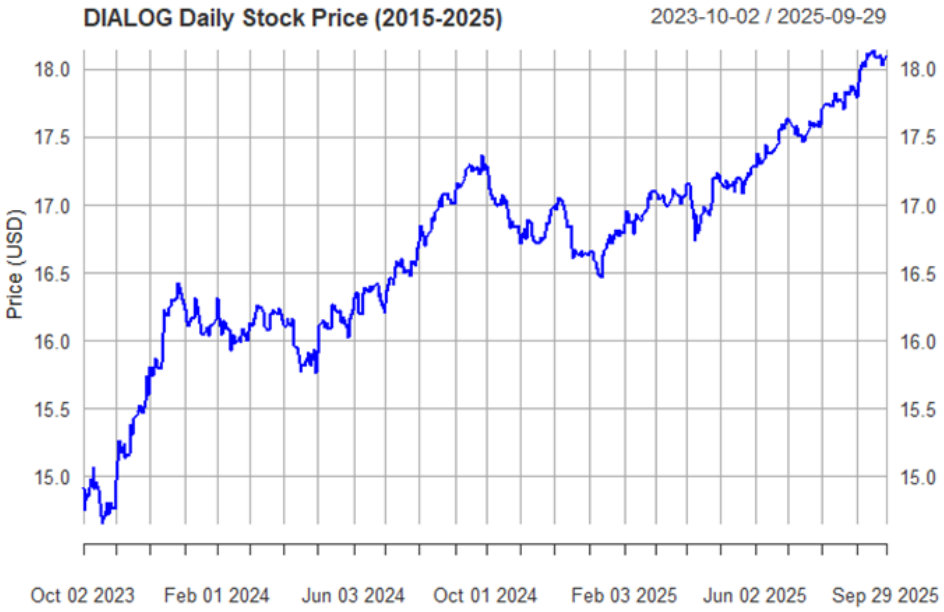


Figure 4.2: Time Series Plot of DIAL.

CHAPTER 5
Results

5.1 Fundamental Analysis

5.1.1 Feature Selection Using Correlation Analysis

A Pearson correlation analysis was conducted to examine linear relationships among features. Independent variables exhibiting very low correlations with other features were removed to reduce noise and redundancy.

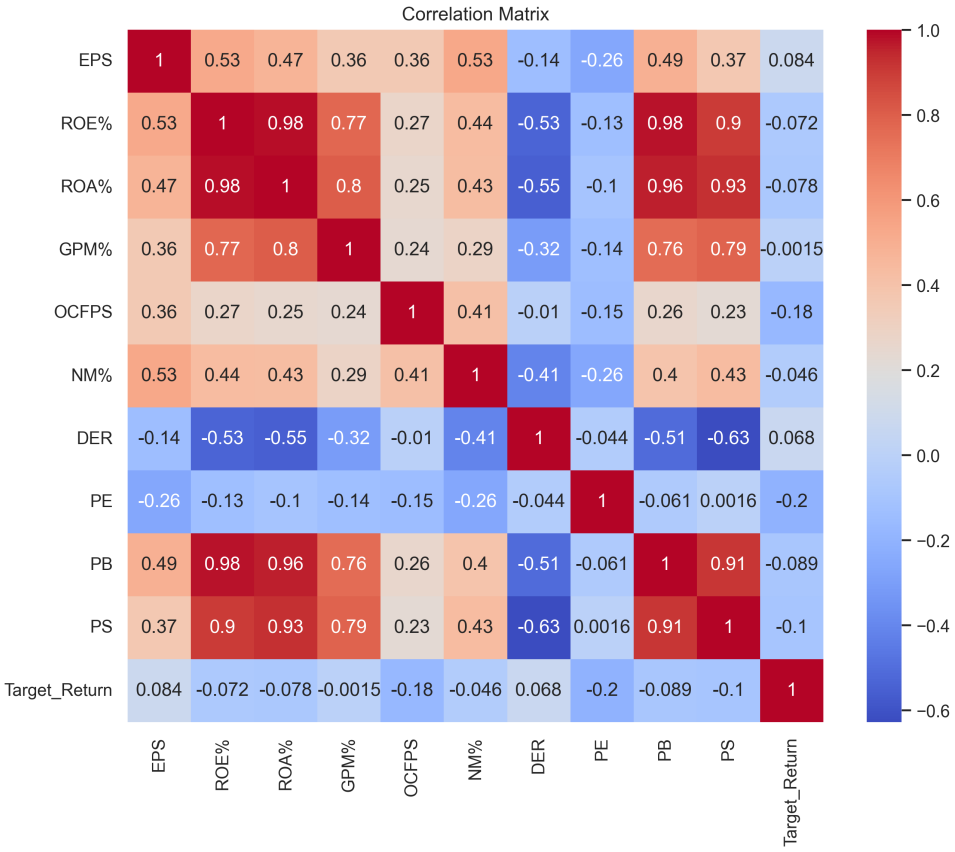


Figure 5.1: Correlation matrix of all fundamental variables.

- According to The correlation plot following weakly correlated variables were removed:
  - PE
  - OCFPS
  - NM%
  - EPS

- The remaining features exhibit stronger relationships, providing a cleaner input set for subsequent machine learning models and improving interpretability while reducing noise.

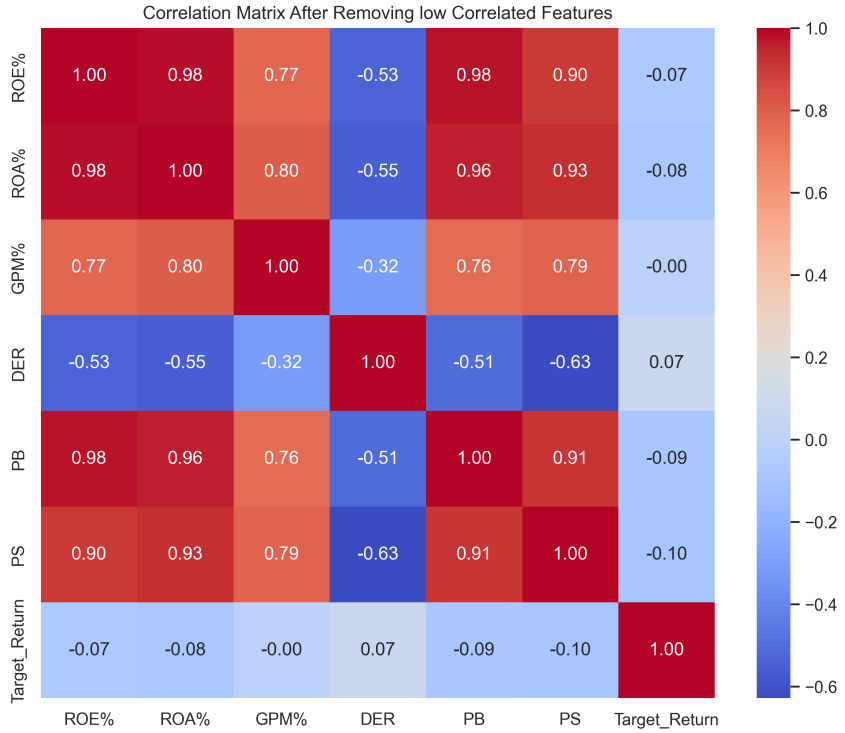


Figure 5.2: Correlation matrix after removing weakly correlated variables.

5.1.2 Classification Model Performance

Model	Average CV Accuracy (%)	Standard Deviation
Random Forest	72.92	0.03
K-Nearest Neighbors (KNN)	70.83	0.03
Support Vector Classifier (SVC)	68.75	0.05
Decision Tree	64.58	0.03
Logistic Regression	62.50	0.135

Random Forest achieved the highest accuracy (72.92%).

Random Forest Probability-Based Ranking

Using the Random Forest classifier, the average predicted probabilities of gains were used to rank companies by their likelihood of positive returns. This visualizes the resulting company rankings, with higher bars indicating higher predicted probabilities of gain.

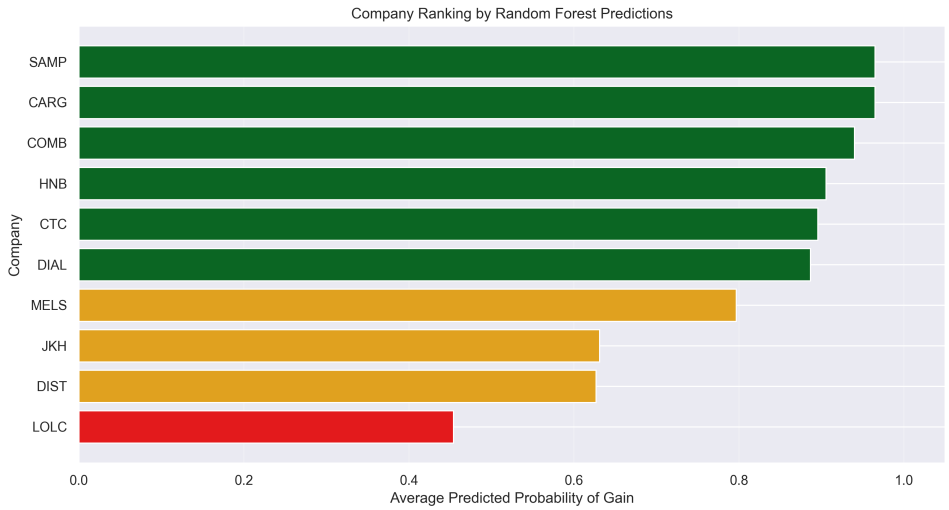


Figure 5.3: Company rankings based on average predicted probability of gain from the Random Forest model.

The performance measurements of the ranking are summarized as follows:

- Mean Rank IC: 0.330968
- Top-3 Hit Rate: 0.375
- IC-IR: 1.053531

5.1.3 Regression Model Performance

Regression models were evaluated using cross-validated R^2 scores. All regression models produced negative average $\mathbf{R^2}$ values, indicating limited explanatory power under cross-validation.

Table 5.1: Regression Model Performance (Cross-Validation R^2 Scores)

Model	Average CV R^2	Standard Deviation
Random Forest Regressor	-0.341	0.355
Support Vector Regressor (SVR)	-0.257	0.164
XGBoost Regressor	-0.181	0.249

5.1.4 Ranking Models

LightGBM Ranker and XGBoost Ranker models were trained to generate company rankings based on the selected features.

Label	Name	LightGBM Score	LightGBM Rank	XGBoost Score	XGBoost Rank
0	CARG	-0.0568	4	-0.1005	5
1	CTC	-0.4007	7	-0.0257	4
2	DIAL	-0.0404	3	0.7255	1
3	DIST	0.0070	2	0.1645	3
4	MELS	-1.5814	10	-0.2407	6
5	JKH	-1.2669	8	-2.3218	10
6	LOLC	-1.4557	9	-0.5496	7
7	COMB	-0.3206	6	-0.6443	9
8	HNB	0.1992	1	0.2973	2
9	SAMP	-0.2271	5	-0.5695	8

Figure 5.4: Comparison of company rankings generated by LightGBM and XGBoost models.

.

Table 5.2: Performance metrics for LightGBM and XGBoost ranking models

Model	Mean Rank IC	Top-3 Hit Rate	IC-IR
LightGBM	0.7554	0.81	55.1868
XGBoost	0.7936	0.746	4.2918

5.1.5 Comparison of All Ranking Approaches

The three ranking approaches XGBoost classifier probability based ranking, LightGBM Ranker, and XGBoost Ranker were compared under the combined feature set.

Table 5.3: Performance metrics for Random Forest, LightGBM, and XGBoost ranking models.

Model	**Mean Rank IC**	**Top-3 Hit Rate**	**IC-IR**
Random Forest	0.3310	0.375	1.0535
LightGBM	0.3557	0.5	1.2057
XGBoost	0.5881	0.625	5.7514

5.1.6 Feature Importance Analysis

The importance of features in predicting company rankings was analyzed for all three models. Figures 5.5 show the feature importance scores for Random Forest, LightGBM, and XGBoost, respectively.

Feature	LightGBM Gain (%)	XGBoost Ranker Gain (%)	Random Forest (%)
ROE%	26.53%	13.02%	16.26%
PS	31.21%	9.79%	13.56%
DER	22.67%	10.86%	17.07%
ROA%	5.92%	27.38%	12.66%
GPM%	6.00%	21.95%	15.71%
PB	7.66%	17.00%	9.66%

Figure 5.5: Feature importance values for Random Forest, LightGBM, and XGBoost models.

5.2 Technical Analysis

5.2.1 Classification Model Performance

Several classification algorithms were evaluated using cross-validation to predict high future returns. Performance was measured by AUC scores.

Table 5.4: Classification model performance (AUC scores)

Model	AUC	Standard Deviation
XGBoost	0.7933	0.0521
Random Forest	0.7912	0.0593
Logistic Regression	0.7846	0.0684

XGBoost achieved the highest AUC (0.7933 ± 0.0521).

XGBoost Classifier Probability-Based Ranking

Companies were ranked according to the average predicted probability of high future returns from the XGBoost classifier.

The ranking performance metrics are:

- Mean Rank IC: 0.7411
- Top-3 Hit Rate: 0.292
- IC-IR: 28.8658

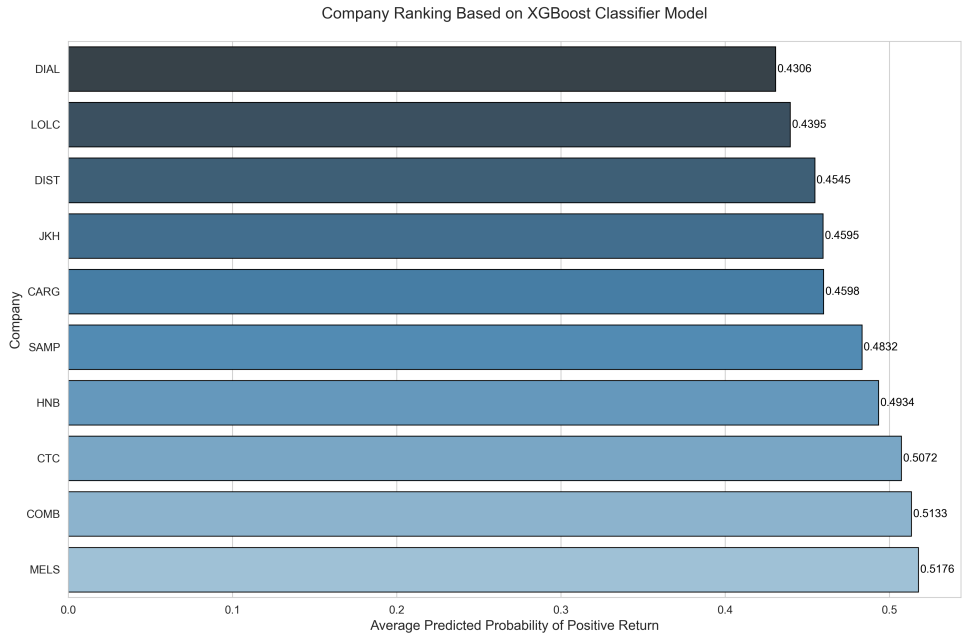


Figure 5.6: Company rankings based on average predicted probability

5.2.2 Regression Model Performance

Regression models were evaluated using cross-validated R^2 scores.

Table 5.5: Regression model performance (Cross-Validation R^2 Scores)

Model	**Average CV R^2**	**Standard Deviation**
Random Forest Regressor	0.2832	0.1383
XGBoost Regressor	0.2311	0.1123

Random Forest Regressor showed the highest (though moderate) explanatory power.

5.2.3 Ranking Models

LightGBM and XGBoost rankers were trained to rank companies by predicted performance. The resulting rankings are shown in Figures 5.7 and 5.8. Both models produce meaningful rankings, though they differ in the relative positions assigned to individual companies.

Company	LightGBM_Score	Overall_Rank
DIAL	-0.6908	1
CARG	-0.6924	2
MELS	-0.7191	3
DIST	-0.8018	4
CTC	-0.8708	5
COMB	-0.897	6
HNB	-0.9112	7
SAMP	-0.9287	8
JKH	-0.9993	9
LOLC	-1.0297	10

Figure 5.7: LightGBM Ranker company rankings.

Company	XGBr_Score	Overall_Rank
DIAL	0.0733	1
HNB	0.0720	2
COMB	0.0609	3
SAMP	0.0590	4
MELS	0.0510	5
CARG	0.0415	6
CTC	0.0339	7
DIST	0.0304	8
JKH	-0.0028	9
LOLC	-0.0221	10

Figure 5.8: XGBoost Ranker company rankings.

Performance metrics for the ranking models are summarized in Table 5.6.

Table 5.6: Performance metrics for LightGBM and XGBoost ranking models

Model	Mean Rank IC	Top-3 Hit Rate	IC-IR
LightGBM	0.7492	0.803	4.8445
XGBoost	0.7844	0.739	4.1291

5.2.4 Comparison of All Ranking Approaches

The three ranking approaches XGBoost Classifier probability ranking, LightGBM Ranker, and XGBoost Ranker were compared. Figure 5.9 shows the ranks assigned by each method.

Company	XGB Classifier Score	XGB Classifier Rank	LightGBM Score	LightGBM Rank	XGBoost Ranker Score	XGBoost Ranker Rank
MELS	0.5176	1	-0.7191	3	0.0510	5
COMB	0.5133	2	-0.8970	6	0.0609	3
DIAL	0.4306	10	-0.6908	1	0.0733	1
HNB	0.4934	4	-0.9112	7	0.0720	2
CARG	0.4598	6	-0.6924	2	0.0415	6
CTC	0.5072	3	-0.8708	5	0.0339	7
SAMP	0.4832	5	-0.9287	8	0.0590	4
DIST	0.4545	8	-0.8018	4	0.0304	8
JKH	0.4595	7	-0.9993	9	-0.0028	9
LOLC	0.4395	9	-1.0297	10	-0.0221	10

Figure 5.9: Company ranks across XGBoost Classifier, LightGBM Ranker, and XGBoost Ranker.

XGBoost Ranker achieved the highest IC (0.7844), while LightGBM Ranker obtained the highest Top-3 Hit Rate (0.803). In comparison, the XGBoost classifier, although not explicitly trained for ranking, still demonstrated competitive ranking performance, achieving a Rank IC of 0.7411, and a Top-3 Hit Rate of 0.292.

5.3 Combined Analysis

In the combined analysis, fundamental indicators and technical indicators were integrated into a single feature set. All classification, regression, and ranking models were trained using this combined features.

5.3.1 Classification Model Performance

Several classification models were evaluated using cross-validation, and performance was assessed using AUC.

Table 5.7: Classification model performance(AUC scores)

Model	AUC	Standard Deviation
XGBoost	0.7986	0.0389
Random Forest	0.7916	0.0457
Logistic Regression	0.7707	0.0637

XGBoost achieved the highest AUC (0.7986 ± 0.0389), indicating improved classification performance when fundamental and technical features are combined.

XGBoost Classifier Probability-Based Ranking

Companies were ranked according to the average predicted probability of high future returns obtained from the XGBoost classifier trained on the combined feature set.

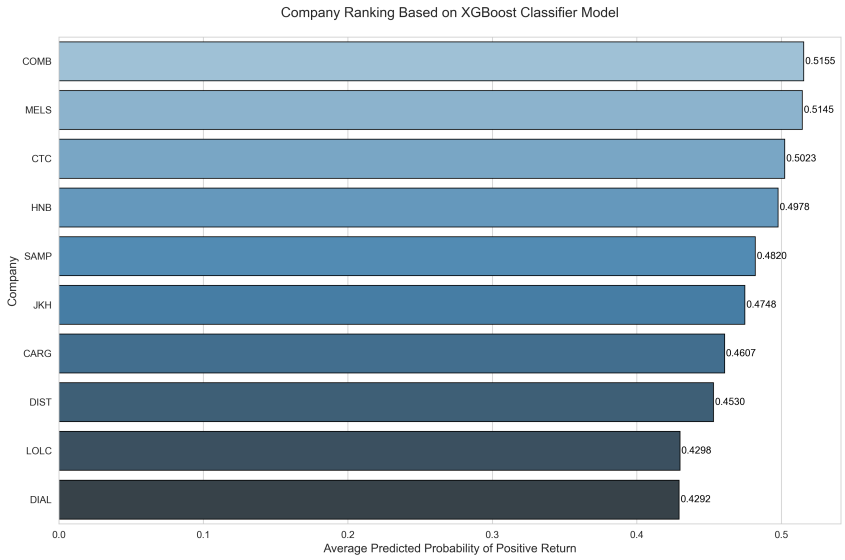


Figure 5.10: Company rankings

The ranking performance metrics are summarized as follows:

- Mean Rank IC: 0.7490
- Top-3 Hit Rate: 0.292
- IC-IR: 46.9370

5.3.2 Regression Model Performance

Regression models were evaluated using cross-validated R^2 scores to assess their ability to predict future returns under the combined feature representation.

Table 5.8: Regression model performance

Model	**Average CV R^2**	**Standard Deviation**
Random Forest Regressor	0.2841	0.1392
XGBoost Regressor	0.2122	0.1147

5.3.3 Ranking Models

Learning-to-rank models were trained to directly rank companies based on predicted performance using the combined fundamental and technical feature set. The company rankings produced by LightGBM Ranker and XGBoost Ranker are shown in Figures 5.11 and 5.12, respectively.

Company	LightGBM_Score	Overall_Rank
CARG	-0.6681	1
DIAL	-0.6915	2
MELS	-0.7247	3
DIST	-0.75	4
CTC	-0.834	5
SAMP	-0.9454	6
HNB	-0.9482	7
COMB	-1.0074	8
JKH	-1.0632	9
LOLC	-1.0775	10

Figure 5.11: LightGBM Ranker company rankings

Company	XGBr_Score	Overall_Rank
DIAL	0.0888	1
HNB	0.0824	2
SAMP	0.0770	3
COMB	0.0750	4
CARG	0.0611	5
MELS	0.0402	6
CTC	0.0399	7
DIST	0.0301	8
JKH	0.0003	9
LOLC	-0.0382	10

Figure 5.12: XGBoost Ranker company rankings

Performance metrics for the ranking models are reported in Table 5.9.

Table 5.9: Performance metrics for LightGBM and XGBoost ranking models

Model	**Mean Rank IC**	**Top-3 Hit Rate**	**IC-IR**
LightGBM	0.7554	0.815	5.1868
XGBoost	0.7936	0.746	4.2918

5.3.4 Comparison of All Ranking Approaches

The three ranking approaches XGBoost classifier probability based ranking, LightGBM Ranker, and XGBoost Ranker were compared under the combined feature set. Figure 5.13 illustrates the company ranks assigned by each method.

Company	XGB Classifier Score	XGB Classifier Rank	LightGBM Score	LightGBM Rank	XGBoost Ranker Score	XGBoost Ranker Rank
MELS	0.5145	2	-0.7247	3	0.0402	6
COMB	0.5155	1	-1.0074	8	0.0750	4
HNB	0.4978	4	-0.9482	7	0.0824	2
CARG	0.4607	7	-0.6681	1	0.0611	5
DIAL	0.4292	10	-0.6915	2	0.0888	1
SAMP	0.4820	5	-0.9454	6	0.0770	3
CTC	0.5023	3	-0.8340	5	0.0399	7
DIST	0.4530	8	-0.7500	4	0.0301	8
JKH	0.4748	6	-1.0632	9	0.0003	9
LOLC	0.4298	9	-1.0775	10	-0.0382	10

Figure 5.13: Company ranks across XGBoost Classifier, LightGBM Ranker, and XGBoost Ranker.

The XGBoost Ranker achieved the highest Rank IC (0.7936), while the LightGBM Ranker obtained the highest Top-3 Hit Rate (0.815).

5.3.5 Feature Importance Analysis

Feature importance was examined to identify the most influential fundamental and technical indicators contributing to model performance under the combined analysis.

Feature	XGB Classifier Weight	LightGBM Gain (%)	XGBoost Ranker Weight
Daily_Return	26.6600	38.6500%	13.0700
ROA%	7.3800	2.5900%	6.6700
PB	5.7000	2.1800%	6.6700
ROE%	7.0700	2.3300%	5.9600
Beta	4.2600	10.5600%	5.9200
GPM%	6.2300	2.1200%	5.8600
PE	4.7600	1.8300%	6.0900
OCFPS	4.7500	3.5700%	5.5100
RSI_(14_days)	4.1900	9.4500%	5.4900
Daily_variance	3.2700	9.7300%	5.6700
EPS	4.6600	1.8000%	5.9000
SL20_index	4.0200	6.2100%	5.3500
DER	3.6800	1.6800%	6.3000
PS	5.0500	1.2900%	5.3900
MA_50	3.9500	4.6000%	5.0100
NM%	4.3600	1.4200%	5.1400

Figure 5.14: Combined feature importance across models

The results indicate that both technical momentum indicators and fundamental valuation measures play complementary roles in improving stock classification and ranking performance.

5.3.6 Stock Selection Rationale

DIAL consistently performed well across the various machine learning models. Although the classification models selected a different best stock than the XGBoost Ranker model, which achieved the highest value of the Rank Information Coefficient, the XGBoost Ranker model selected DIAL as the number one ranked stock. As the best performing model, the results of the XGBoost Ranker were used to select DIAL for further time series analysis and short-term price forecasting.

5.3.7 Price Behavior and Preliminary Analysis

Time series plot illustrates the daily adjusted closing prices of DIAL from October 2023 to September 2025. The series exhibits a clear upward trend with short-term fluctuations, suggesting potential non-stationarity in the level series.

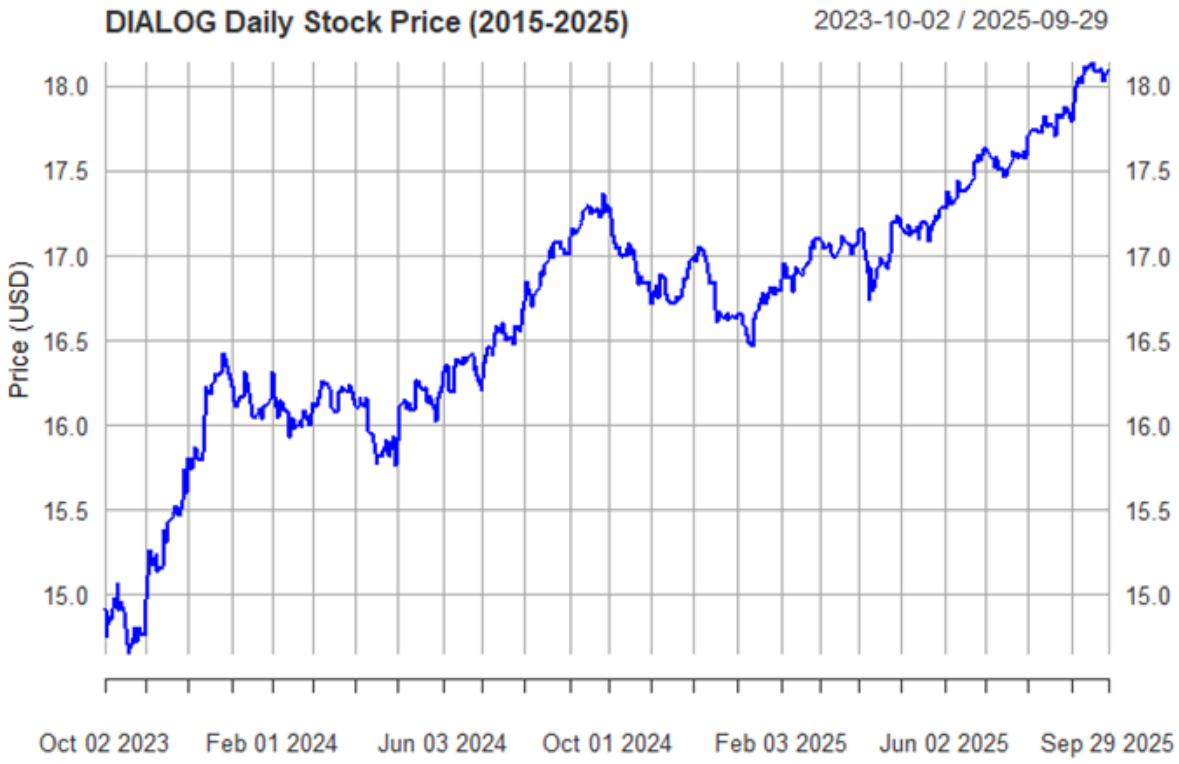


Figure 5.15: Time series plot of DIAL stock

To further assess the autocorrelation structure of the raw series, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots are shown in Figure 5.16. The slowly decaying ACF confirms the presence of a trend and non-stationarity, while the significant spike in the PACF at lag 1 is typical of non-stationary financial time series.

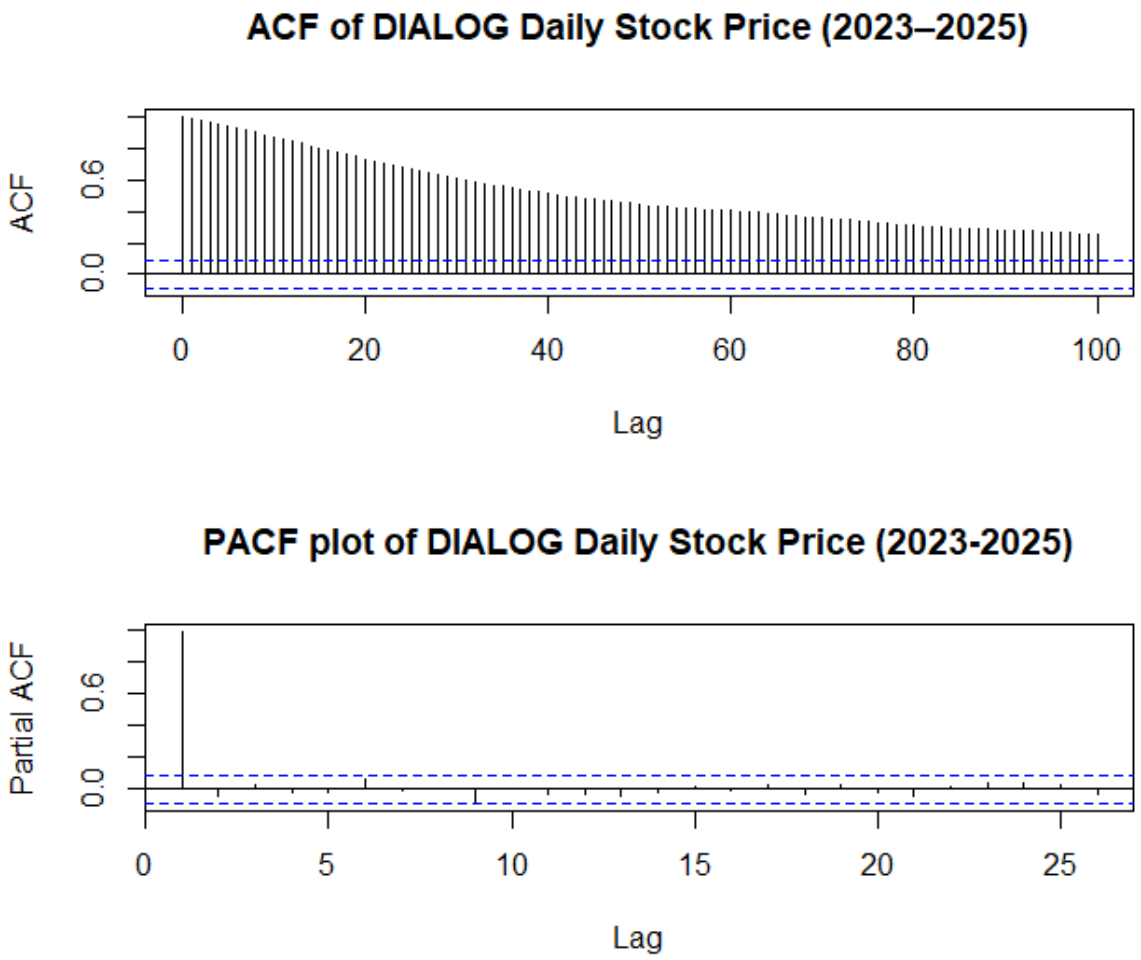


Figure 5.16: Time series plot of DIAL stock

5.3.8 Stationarity Testing

The KPSS test and ADF test was applied to examine the stationarity of the DIAL price series, for the raw price series at the 5% significance level.

We can see that the series is Non- Stationary using staticaly tests.

Augmented Dickey–Fuller (ADF) Test

The Augmented Dickey–Fuller (ADF) test is used to examine the presence of a unit root in a time series. The existence of a unit root indicates that the series is non-stationary.

- **Null Hypothesis (H_0):** The time series has a unit root and is non-stationary.
- **Alternative Hypothesis (H_A):** The time series does not have a unit root and is stationary.

Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Test

The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test is used to assess the stationarity of a time series around a deterministic trend.

- **Null Hypothesis (H_0):** The time series is stationary around a deterministic trend (trend-stationary).
- **Alternative Hypothesis (H_A):** The time series is non-stationary and contains a unit root.

Table 5.10: Stationarity test results For Row price series

Test	p-value	Conclusion
KPSS Test	0.01	Non-Stationary
ADF Test	0.4105	Non-Stationary

To achieve stationarity, first-order differencing was applied to the row prices.

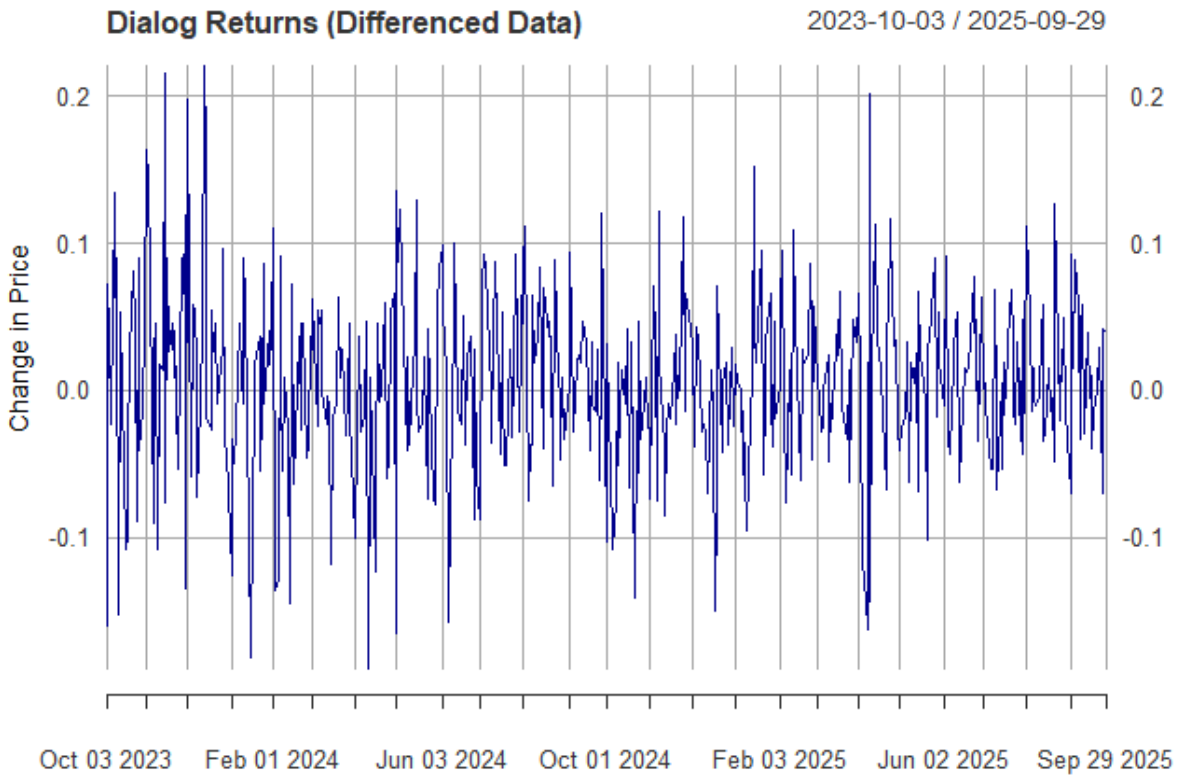


Figure 5.17: First-order differenced series of DIAL adjusted closing prices.

Figure 5.17 displays the first-order differenced series, which fluctuates around a mean of zero with no apparent trend

Then got the stationarity conclusion from the KPSS test and ADF test.

Table 5.11: Stationarity test results For First-order differenced price series

Test	p-value	Conclusion
KPSS Test	0.1	Stationary
ADF Test	0.01	Stationary

5.3.9 Model Identification via ACF and PACF

To identify appropriate ARIMA orders, the ACF and PACF plots of the stationary (differenced) series were examined (Figure 5.18).

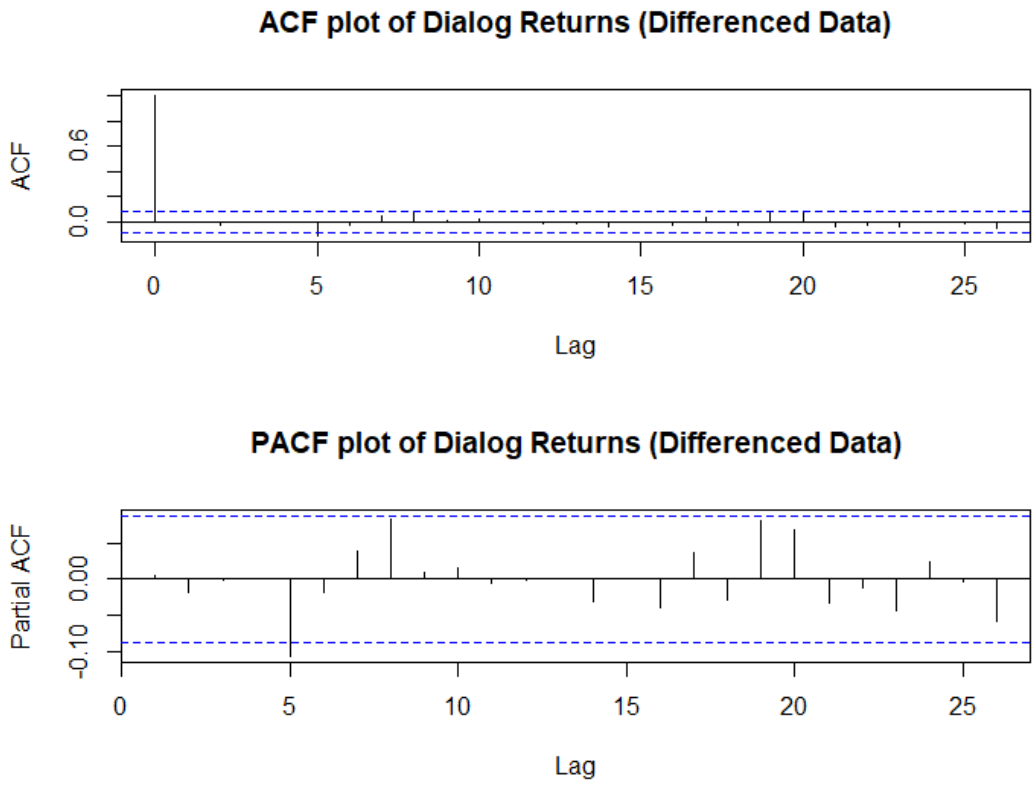


Figure 5.18: ACF and PACF plots for the first-order differenced DIAL price series.

5.3.10 ARIMA Model Selection

Several ARIMA specifications were estimated and compared using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Table 5.12 summarizes the results.

Table 5.12: ARIMA model comparison for DIAL

Model	**AIC**	**BIC**
ARIMA(0,1,0)	-1024.618	-1020.407
ARIMA(5,1,0)	-1264.990	-1239.726
ARIMA(0,1,5)	-1347.332	-1322.068
ARIMA(1,1,1)	-1353.178	-1340.546
Auto ARIMA (3,1,1)	**-1358.102**	-1332.826

The Auto ARIMA model with specification ARIMA(3,1,1) achieved the lowest AIC and was therefore selected for forecasting.

The summary details of the ARIMA(3,1,1) model as follows,

Table 5.13: Training Set Error Measures

	ME	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**
Training set	0.000118	0.0613	0.0467	0.0004	0.2839	0.988	-0.00147

The ARIMA model is based on the assumptions that is the residuals (errors) are follow a normal distribution and the residuals should have constant variance over time. The normality assumption can be checked using a Q-Q plot .

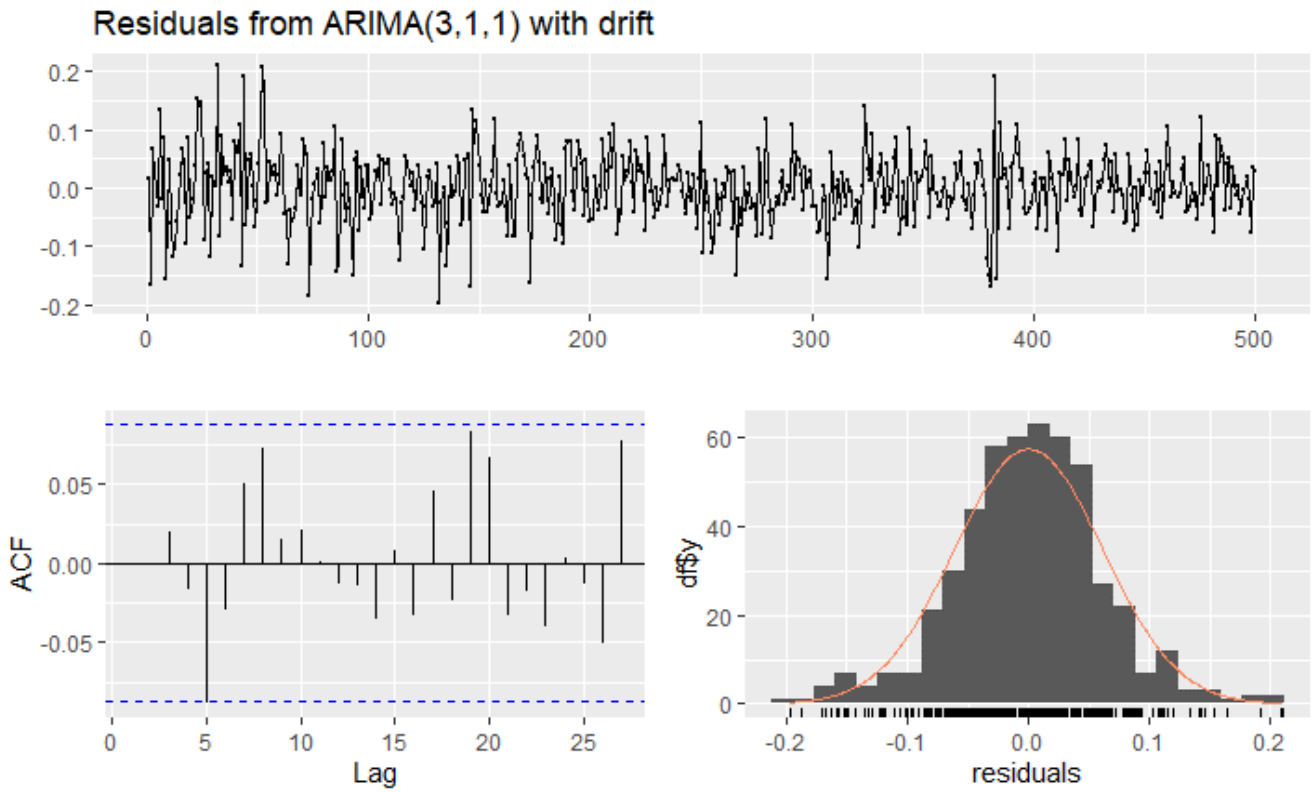


Figure 5.19: ARIMA(3,1,1) model residuals

Ljung-Box Test Result

```
        Box-Ljung test

data:  residuals(fit)
X-squared = 17.884, df = 20, p-value = 0.5951
```

Ljung-Box test was conducted on the model residuals to assess autocorrelation.the results indicate no significant autocorrelation in the residuals.(p value>0.05) This confirms that the model adequately fits the data and the residuals behave like white noise.

5.3.11 Thirty-Day Price Forecast

Using the selected ARIMA(3,1,1) model, the daily adjusted closing price of DIAL was forecasted for the next 30 trading days. Figure 5.21 presents the point forecasts along with 80% and 95% confidence intervals.

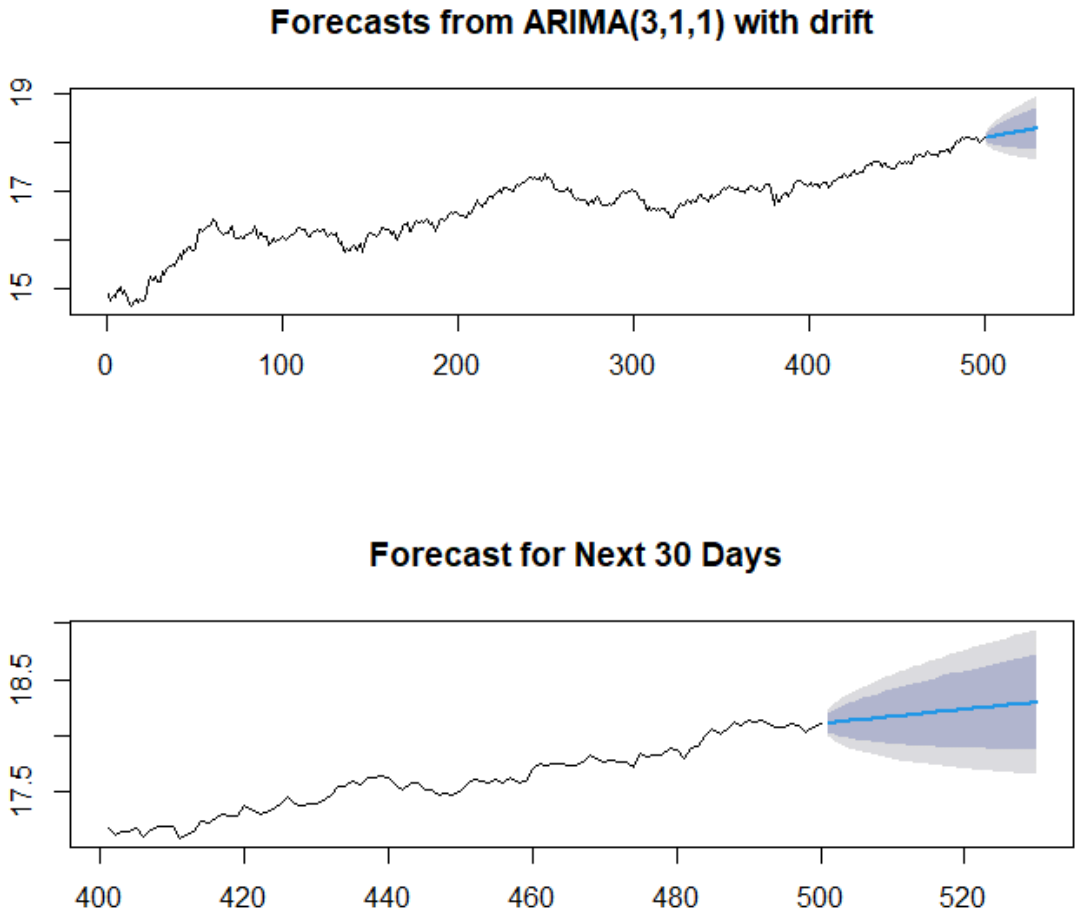


Figure 5.20: Thirty-day forecast of DIAL stock price with confidence intervals.

5.3.12 Forecast Interpretation

The forecast indicates a gradual upward movement in the expected price of DIAL over the next 30 trading days. The relatively narrow confidence intervals suggest moderate forecast uncertainty, reflecting stable short-term price dynamics. These results support the findings of the machine learning analysis, reinforcing DIAL's position as a strong candidate for short-term investment consideration.

Day	Forecast	Lo_95	Hi_95
1	18.11419	17.99340	18.23497
2	18.11864	17.94725	18.29003
3	18.12524	17.91648	18.33400
4	18.13149	17.89255	18.37044
5	18.13807	17.87133	18.40481
6	18.14431	17.85309	18.43554
7	18.15083	17.83649	18.46516
8	18.15712	17.82168	18.49256
9	18.16359	17.80798	18.51921
10	18.16992	17.79546	18.54437
11	18.17637	17.78377	18.56896
12	18.18271	17.77294	18.59248
13	18.18914	17.76276	18.61552
14	18.19550	17.75324	18.63777
15	18.20192	17.74424	18.65960
16	18.20829	17.73576	18.68082
17	18.21470	17.72772	18.70168
18	18.22108	17.72011	18.72205
19	18.22748	17.71287	18.74210
20	18.23387	17.70599	18.76174
21	18.24027	17.69943	18.78110
22	18.24665	17.69318	18.80013
23	18.25305	17.68720	18.81890
24	18.25944	17.68150	18.83738
25	18.26583	17.67603	18.85563
26	18.27222	17.67081	18.87364
27	18.27862	17.66580	18.89143
28	18.28501	17.66100	18.90901
29	18.29140	17.65640	18.92640
30	18.29779	17.65199	18.94359

Figure 5.21: Thirty-day forecasted values of DIAL stock price with 95% confidence intervals.

CHAPTER 6
Discussion and Conclusion

6.1 Discussion

In this this study developed Machine Learning–ARIMA framework for price forecasting and stock selection in the Colombo Stock Exchange's (CSE) SL20 Index.

In the first stage, Machine Learning techniques including classification, regression, and ranking models are employed to develop 3 analysis are fundamental, technical, and combined Analysis to identify the best-performing company among selected SL20 firms.

The fundamental analysis revealed that classification models, particularly Random Forest (72.92% accuracy), performed reasonably well. However, regression models produced negative R^2 values, indicating limited predictive power when using fundamental indicators alone.

The technical analysis demonstrated substantially stronger performance. The XGBoost classifier achieved an AUC of 0.7933, while ranking models exhibited dramatically higher Rank IC values (LightGBM: 0.7492, XGBoost: 0.7844) with the LightGBM Ranker achieving a Top-3 Hit Rate of 0.803.

The combined analysis gives the best results.The combine of both fundamental, technical indicators provides the best-performing model for forecasting the next 30 days of closing prices for the selected stock.The XGBoost Ranker achieved the highest Rank IC (0.7936), while the LightGBM Ranker obtained the highest Top-3 Hit Rate (0.815), where Dialog Axiata PLC (DIAL) selected as the top-performing stock.

Feature importance analysis gave that variables such as ROE, P/S ratio (price relative to sales), and D/E ratio (debt compared to equity) had the biggest effect on predicting stock performance. In the second stage time series techniques are used to analyze the future price behavior of the chosen stock.

In the second stage time series techniques are used to analyze the future price behavior of the chosen stock. To achieve stationarity, Stationarity testing revealed non-stationarity in the raw price series. first-order differencing was applied to the row prices. After testing the stationarity conclusion from the KPSS test and ADF test with confirmed by visual

inspection and ACF/PACF plots, we conclude that. First-order differencing transformed the series to stationarity

The Auto ARIMA model, ARIMA (3,1,1) achieved the lowest AIC and was selected to predict DIAL's short-term price movements. 30-day forecast shows a gradual upward price trend with moderate uncertainty

6.2 Conclusion

This study demonstrates that a two-stage hybrid framework combining machine learning and ARIMA can effectively support investment decisions in the Sri Lankan market. By filling in the gaps in the Colombo Stock Exchange's systematic methodologies, the framework enhances market efficiency and decision-making. After forecasting DIAL's short-term price movements,the 30-day prediction showed a stable upward trend with moderate uncertainty.

Overall, this study demonstrates that integrating ARIMA forecasting with machine learning stock selection results in a reliable, data-driven tool to assist investors in the Sri Lankan stock market.For future research, this framework could be improved by trying other machine learning algorithms, using more detailed data, making longer-term forecasts, or including additional financial and market indicators to make predictions more accurate and useful for investors.

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APPENDIX A
Supplementary Materials

This appendix provides access to the data and source code used in this study.

- **Dataset (Google Sheets):** Access the dataset
- **Python Code (Google Drive Folder):** Access the source code

APPENDIX B
R codes For Time Series Forecasting

```
# ============================================================
# STEP 1: INSTALL AND LOAD PACKAGES
# ============================================================
# If you haven't installed these yet, run the install lines once:
if(!require(quantmod)) install.packages("quantmod")
if(!require(tseries)) install.packages("tseries")
if(!require(forecast)) install.packages("forecast")

library(quantmod) # For downloading stock data
library(tseries)  # For ADF Stationarity Test
library(forecast) # For ARIMA modeling

# ============================================================
# STEP 2: DOWNLOAD DATA (2015 - 2025)
# ============================================================
# We fetch Unilever (UL) directly from Yahoo Finance
# This ensures the dates are formatted perfectly
getSymbols("DIAL", src = "yahoo", from = "2023-10-01", to ="2025-09-30")
head(DIAL)
# Select the 'Adjusted Close' column (Column 6)
# We use Adjusted Close to account for dividends
stock_data <- DIAL[,6]
head(stock_data)
# Check if data loaded correctly
head(stock_data)

plot(stock_data, main="DIALOG Daily Stock Price (2023-2025)",
     ylab="Price (USD)", col="darkblue")

par(mfrow=c(2,1))
acf(stock_data,
    lag.max = 100,
```

```
    main = "ACF of DIALOG Daily Stock Price (2023  2025  )")
acf(stock_data, main="ACF plot of DIALOG Daily Stock Price (2023-2025)")
#not stationary
pacf(stock_data,main="PACF plot of DIALOG Daily Stock Price (2023-2025)"
   )

# ========================================================
# STEP 3: STATIONARITY TEST (KPSS TEST)
# ========================================================
print("--- KPSS TEST RESULTS (RAW DATA) ---")
# Null Hypothesis: Data is Stationary
kpss.test(stock_data)
# If p-value < 0.05, we reject H0
adf.test(stock_data,alternative = "stationary")


stock_diff1 <- diff(stock_data)
stock_diff1 <- na.omit(stock_diff1) # Remove the first NA value created
   by differencing

plot(stock_diff1, main="Dialog Returns (Differenced Data)",
     ylab="Change in Price", col="darkgreen",lwd=1.5)

par(mfrow=c(2,1))
acf(stock_diff1, main="ACF plot of Dialog Returns (Differenced Data)")
pacf(stock_diff1, main="PACF plot of Dialog Returns (Differenced Data)")


print("--- KPSS TEST RESULTS (DIFFERENCED DATA) ---")
kpss.test(stock_diff1)

adf.test(na.omit(stock_diff1), alternative = "stationary")

#p-value = 0.1 > 0.05----we cannot reject Ho----stationary

```

```r
checkresiduals(stock_diff1)

#####------Models------#####

Model1<-arima(stock_diff1, order = c(0,1,0))
Model2<-arima(stock_diff1, order = c(1,1,0))
Model3<-arima(stock_diff1, order = c(0,1,1))
Model4<-arima(stock_diff1, order = c(1,1,1))
Model5<-arima(stock_diff1, order = c(5,1,0))
Model6<-auto.arima(stock_data)
fit <- auto.arima(stock_data)
accuracy(fit)
resid<- fit$residuals
plot(resid,ylab="residuals",col="darkblue")

checkresiduals(fit)

######Compare the models###############

# Model names
model_names <- c("Model1","Model2","Model3","Model4","Model5","Model6")

# Extract AIC and BIC values
aic_vals <- AIC(Model1,Model2,Model3,Model4,Model5,Model6)$AIC
bic_vals <- BIC(Model1,Model2,Model3,Model4,Model5,Model6)$BIC

# Create final data frame
Model_compare <- data.frame(
  Model = model_names,
  AIC_Value = aic_vals,
  BIC_Value = bic_vals
)

Model_compare

```

```
#############Select the best model#####################

Model_compare$Model[which.min(Model_compare$AIC_Value)]
Model_compare$Model[which.min(Model_compare$BIC_Value)]

#For forecasting (prediction)     AIC is preferred
#For model explanation / simplicity     BIC is preferred

#So we get best model as Model6

Best_model<-Model6
Best_model

#############---------FORECASTING-------#########################

forecast_Model6 <- forecast(Model6, h = 30)
forecast_Model6

plot(forecast_Model6)


###Residuals are follow WhiteNoice##########

forecast_Model6$fitted
plot(forecast_Model6$fitted)

checkresiduals(forecast_Model6$residuals)


###### Forecasted values ##########

forecast_Model6 <- forecast(Model6, h = 30)
par(mfrow=c(2,1))

plot(forecast_Model6)
```

```
136
137 plot(forecast_Model6,
138      include = 100,   # shows only last 100 observations
139      main = " Forecast for Next 30 Days")
140
141
142 forecast_Model6$mean
143 as.numeric(forecast_Model6$mean)
144
145
146 data.frame(
147   Day = 1:30,
148   Forecast = forecast_Model6$mean,
149   Lo_80 = forecast_Model6$lower[,1],
150   Hi_80 = forecast_Model6$upper[,1],
151   Lo_95 = forecast_Model6$lower[,2],
152   Hi_95 = forecast_Model6$upper[,2]
153 )
154
155
156 data.frame(
157   Day = 1:30,
158   Forecast = forecast_Model6$mean,
159   Lo_95 = forecast_Model6$lower[,2],
160   Hi_95 = forecast_Model6$upper[,2]
161 )
162
163 ### Model Adequacy #####
164
165 library(TSA)
166
167 # Ljung-Box test
168 Box.test(residuals(fit), lag = 20, type = "Ljung-Box")
```